
The Song, Not the Pin: Route Transfer and Meaning-Based Place Identity in Collective Memory

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Abstract

Problem. The companion warp paper measures whether an agent can reach a target known only from a peer’s memory. Two assumptions remain hidden in that construction — and in essentially all shared-memory multi-agent systems. First, what transfers is a *point*: broadcast snapshots carry place summaries but no edges, so the traveler learns **WHERE** and must re-derive **HOW**. Second, place identity is a *shared coordinate frame*: ‘the same place’ means ‘close (x, y) in a grid the environment hands to every agent’; semantics only breaks ties inside a spatial radius. In the source culture a songline is neither: the song **IS** the route, and its places are identified by what they look like, not by coordinates nobody shares. **Approach.** We remove both assumptions and measure what each was hiding. (i) *Route warp*: broadcasts carry traversed edges with provenance; φ_{route} is the foreign share of edge mass along the committed path, and a strict RW^* is a connected foreign sub-path the traveler has never walked. (ii) *Semantic place identity*: places are matched by local semantic constellations (translation-invariant fingerprints); the inter-agent frame — translation in one step, full $SE(2)$ with rotations in the next — is recovered from mutually-unique matches of co-visited landmarks, and foreign evidence, including never-seen targets, is transported through it. **Findings.** What route knowledge buys is a step function, not a slope: place-only transfer collapses at the **FIRST** wall (success 1.0 at detour $D=1.00$ vs. 0.0 at $D=1.38$, and worse than blind exploration), while route transfer completes everywhere. Aging breaks the transferred route at a point predictable to the exact edge (110/110 regimes; all 40 mid-route ruptures across two world classes at the predicted index, error 0), and safety travels with the route: 0 hazard hits over the witness’s path vs. 3.25 [2.45, 4.10] for the place-only traveler, with the risk **RESUMING** at the predicted rupture point. Meaning-based identity recovers secret frames exactly (100% translation; 80/80 under $SE(2)$), restores 88% of the shared-frame performance in the full peer stack, and fails *closed* under perceptual aliasing where the coordinate assumption fails *open* — and the companion distance law re-emerges tick-exactly (breakpoints 20/14/9) with no shared origin and no shared north. **Thesis.** A shared map is not what makes collective navigation work. What transfers is the song — a sequence with meaning-anchored places — and each impoverishment of the song (point instead of route, coordinates instead of meaning) fails in a specific, measurable, and now predicted way. **Scope.** Grid worlds; translation and 90° rotations; measurement and minimal mechanisms, not a general theory.

1 Introduction

A songline encodes a journey: a sequence of places, each recognisable by what is there, bound into a path by the song’s order. When multi-agent systems “share memory,” they typically share something much poorer — a pin on a common map: coordinates of a target in a frame all agents are assumed to inhabit. The companion warp paper¹ made the act of navigating by someone else’s memory a measured event (W^* : a materialisation whose evidence is entirely foreign) and showed its value is governed by the coordination contract attached to the information. But its construction inherited the two impoverishments silently:

1. **The pin, not the song.** Broadcast snapshots carry place concepts — centroids, tags, confidences — but no edges. The receiver learns WHERE the water is; the route is re-derived from scratch. In open grids this is invisible: the manhattan path is the route. The moment paths are expensive, it is everything.
2. **Coordinates, not meaning.** ‘Same place’ is operationalised as coordinate proximity, with semantics as a tie-breaker inside a radius. The cross-agent place-correspondence problem — the reason map merging is hard in robotics — is bracketed by fiat.

This paper removes both assumptions, one at a time, with the measurement discipline of the series: every claim is registered before the runs, failures against registration are reported with their mechanisms, and the strongest results are exact-match validations of closed-form predictions.

Contributions.

1. **Route warp** (RW^* , φ_{route}): edge-level provenance and the strict route-warp event — a connected foreign sub-path the traveler never walked (§3). The value of route over place transfer is a *cliff*, not a slope: place-only transfer dies at the first wall ($1.0 \rightarrow 0.0$ success between measured detour 1.00 and 1.38) and is WORSE than no transfer at all there, while route transfer completes everywhere (§3.1).
2. **The rupture law:** a witness stamps edges sequentially, so its trace ages non-uniformly and a delayed traveler chases a fading song. Replaying the series’ trust $\times$ staleness gate over calibrated kinematics predicts *where the song breaks* to the exact edge: 110/110 regime matches, all 21 mid-route ruptures at the predicted index (error 0), and a discrete trust-flip that zeroes route commits entirely (§4).
3. **Hazard stratification:** the wall and hazard axes are independent — walls break place-transfer’s *completion*, hazards break its *safety* (0 vs. 3.25 hits at equal completion and +0.7 ticks), and when the song ruptures the risk resumes at the predicted edge (§5).
4. **Meaning-based place identity:** translation-invariant semantic fingerprints, mutually-unique landmark matching, and consensus frame recovery — translation first, full $SE(2)$ with secret 90° rotations after (§6). Exact frame recovery (100%; 80/80 under $SE(2)$); online alignment in the full peer stack recovering 88% of the shared-frame gap; a proved failure asymmetry (semantic identity fails *closed* under aliasing and under in-principle-unrecoverable 180° symmetry; coordinate identity fails *open*); and the companion distance law reproduced tick-exactly (20/14/9) with no shared frame at all (§6.1).

2 Setting

Grid worlds (12×10 to 14×12) with semantic water targets, hazards, and walls; agents with symbolic place memory, peer broadcast on cadence K , and the series’ standard planner (target lock = the Q/R/M/C M^* event; completion C^*). The trust $\times$ staleness inclusion rule of the collective-memory companion is used throughout with unchanged constants: evidence mass trust $\cdot e^{-\alpha \text{ age}} \cdot m$, threshold $\tau=0.30$, $\alpha=0.05$. Warp events follow the companion warp paper: provenance φ frozen at lock time; completion attributed to a continuous lock from a foreign commit ($\varphi=1$ at first lock) to success.

¹Companion submissions: the Q/R/M/C measurement framework, the collective-semantic-memory architecture study, and the semantic-warp measurement paper. Citations deferred per concurrent-submission policy.

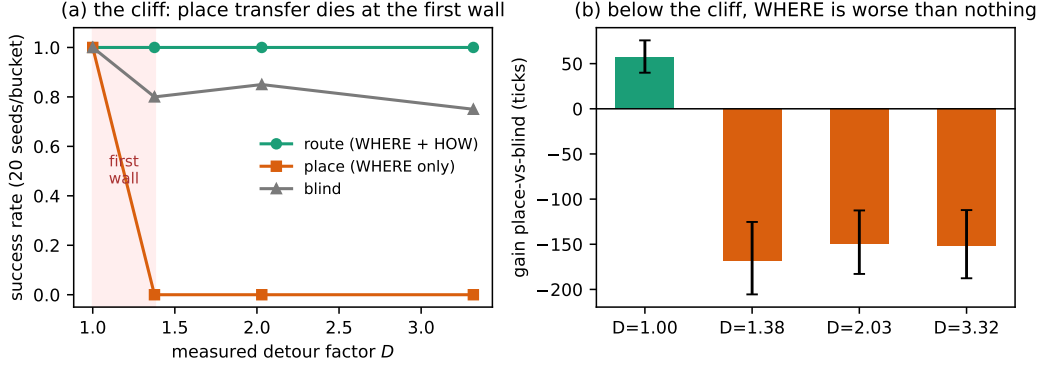


Figure 1: The cliff (paired arms on identical mazes, 20 seeds per measured-detour bucket). **(a)** Place-only transfer does not degrade with detour — it collapses at the FIRST wall, located by the probe bucket between $D=1.00$ and $D=1.38$; route transfer completes everywhere. **(b)** Below the cliff, knowing WHERE without HOW is *worse than knowing nothing*: the known-target manhattan pull pins the planner against walls (−150 to −168 ticks vs. blind exploration), while in the open world the same knowledge is worth +57.

3 Route warp: transferring the HOW

Edge provenance. Broadcast snapshots additionally carry the sender’s OWN traversed edges ($a, b, \text{count}, \text{last tick}$); foreign edges are never re-broadcast (the same privacy contract as place snapshots: no agent reads another’s graph, transport only). For the path P a planner commits,

$$\varphi_{\text{route}}(P) = \frac{\sum_{e \in P, \pi(e) \neq \text{self}} w(e)}{\sum_{e \in P} w(e)}, \quad w(e) = \text{trust} \cdot e^{-\alpha \text{age}(e)} \cdot \log(1 + \text{count}(e)),$$

the same weights the route gate uses. A **strict RW^*** is a committed, traversed, connected sub-path of ≥ 5 edges with $\varphi_{\text{route}} = 1$: the whole verse is someone else’s. Place warp is the degenerate case (path length 0: only the target is foreign), giving a two-dimensional provenance plane ($\varphi_{\text{place}}, \varphi_{\text{route}}$).

First strict RW^* . On a comb maze with measured detour $D=2.65$, a traveler commits and walks a fully foreign 45-edge route to completion in 54 ticks; with the same foreign PLACE evidence but no edges, the standard planner never arrives (300-tick budget), and blind exploration takes 126.

3.1 The cliff: what route knowledge buys

Table 1: Three arms on identical mazes (paired by seed), bucketed by MEASURED detour factor D ; 20 seeds per bucket, censored gains (fail = 300). Place-only transfer does not degrade with D — it collapses at the first wall, and is worse than blind exploration there (the known-target manhattan pull pins the planner against walls). Route transfer completes everywhere; every completion is a strict RW^* with $\varphi_{\text{route}} = 1$.

D	gain route-vs-place [CI]	gain route-vs-blind [CI]	succ r/p/b
1.00	+4.7 [+3.4, +6.0]	+62 [+44, +80]	20/20/20
1.38	+281 [+279, +283]	+113 [+77, +155]	20/0/16
2.03	+273 [+271, +275]	+123 [+91, +160]	20/0/17
3.32	+262 [+259, +264]	+110 [+72, +152]	20/0/15

Figure 1 shows both faces of the result. Two pre-registered predictions failed here, in instructive ways. The negative control at $D=1$ was registered in ticks ($|\text{gain}| \leq 3$) and came out +4.7: the gap is *turn economy* (BFS routes walk straight legs; the dominant-axis planner alternates axes), not route knowledge — in path LENGTH the gap is exactly 0.0 [0.0, 0.0] across all 20 open-grid cells. And “gain grows with D ” is unmeasurable: with place success at 0 the gain saturates at the censoring ceiling. The correct form of the law is a step located by the probe bucket at $D \in (1.00, 1.38)$, i.e. the first wall.

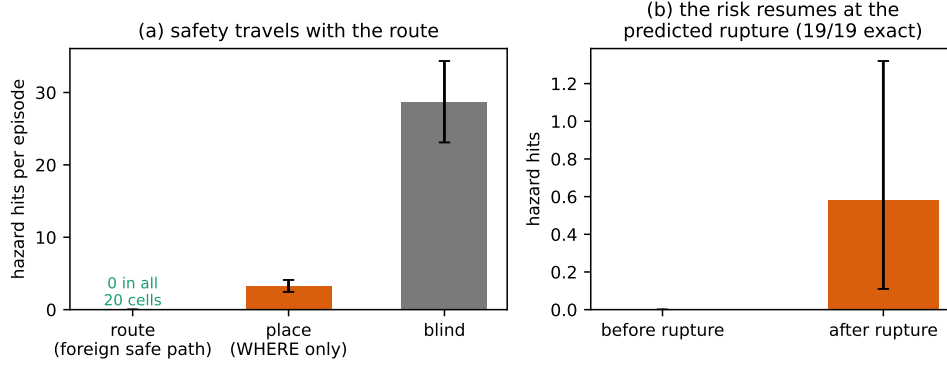


Figure 2: The song protects. **(a)** Hazard hits per episode: the traveler on the witness’s route inherits its safety outright (0 in all 20 worlds); the place-only traveler pays 3.25 hits for the same completion; blind exploration pays 28.7. **(b)** Composition with the rupture law: in the rupture regime the traveler takes zero hits while the song lasts, and the risk resumes after the break — which occurs at the R2-predicted edge index in all 19/19 cells.

100 4 The rupture law: where the song breaks

101 The witness stamps edge i at tick i : its trace ages non-uniformly, the start of the route always
 102 older than the end, and a traveler delayed by a_0 *chases a fading song*. Under the standard gate the
 103 binding edge is the CURRENT one (the oldest remaining), giving three regimes on the a_0 axis:
 104 complete; rupture at a predictable path index; dead on arrival. The predictor is fully deterministic
 105 — a no-decay calibration run records the traveler’s kinematics (path index per tick, turns included),
 106 and the registered prediction replays the closed-form gate over that trajectory, the same device that
 107 produced the companion paper’s 6/6 hold-out.

108 Across 5 mazes $\times$ 11 delays $\times$ 2 trusts (110 cells, registered before execution): **110/110 regime**
 109 **matches; all 21 mid-route ruptures at exactly the predicted edge index (error 0)**; and a route
 110 trust-flip — at trust 0.7 the single-traversal edge mass gives $\text{age}_{\max} = 9.6 < L$, so *no route ever*
 111 *commits* (0 commits observed; the discrete off-switch of the companion law, now for paths). The
 112 rupture of a transferred route is thus not noise but a law: the same constants (α, τ , trust) that set
 113 WHEN a place-warp gate closes set WHERE a route-warp breaks.

114 5 Hazard stratification: the song protects

Table 2: Open fields dense with passable-but-punished hazards (density 0.20), the witness’s route hazard-avoiding by experience; 20 paired worlds, decay off (the safety axis isolated from the aging axis). Both transfer modes COMPLETE — the wall axis (Table 1) and the hazard axis are independent; they differ in the price.

Arm	hazard hits [CI]	completion	mean t
route (foreign safe path)	0 (all 20 cells)	20/20	16.6
place (WHERE only)	3.25 [2.45, 4.10]	20/20	15.9
blind	28.7	—	—

115 The traveler following the witness’s path inherits its safety outright, at a time premium of +0.7
 116 ticks (Figure 2a). Composing with §4: in the rupture regime (19/20 worlds) the traveler takes **0 hits**
 117 **before the rupture** and 0.58 [0.11, 1.32] after it, when it falls back on place knowledge and re-enters
 118 the field — with all 19 ruptures again at the R2-predicted index (error 0; the edge-exact predictor
 119 replicated on a new world class with no adjustment). The song protects while it lasts, and it stops
 120 protecting exactly where the law says the verse ends.

6 Meaning-based place identity

Everything so far — and everything in the companion papers — rides on a shared coordinate frame. We now remove it: each agent’s private frame is the true grid under a secret translation (later: plus a secret 90° -multiple rotation), and broadcasts carry private coordinates that are meaningless to the receiver until it decides what the sender’s places MEAN.

Fingerprints and the handshake. A place fingerprint is the local semantic constellation around a visited cell — which content tags (hazards, walls, water) sit at which relative offsets — translation-invariant and coordinate-free. Matching is *mutually unique* (a pair is accepted only if each side is the other’s sole candidate above an exact-equality threshold; the sender knows far more places than the receiver, and one-directional uniqueness aliases distant look-alikes into the receiver’s neighbourhood), fingerprints need ≥ 2 salient keys, and the frame translation is the modal delta of ≥ 3 agreeing pairs. Landmarks must carry at least one CONTENT tag: border-only (‘void’) patterns describe the shape of the world, not a place, and are excluded — the lesson of the corner trap below.

Witness-traveler results. Over $4 \text{ offsets} \times 2 \text{ hazard densities} \times 10 \text{ seeds}$: the semantic aligner recovers the secret offset *exactly in 100%* of landmark-world runs (mean 15.1 mutual pairs) and completes with strict W^* ($\varphi=1$) every time; the coordinate baseline under any nonzero offset locks a non-water cell in 100% of runs (its rare ‘successes’, 6.7%, are the displaced route accidentally crossing the true water). In the aliased world (featureless interior band) the two identity regimes break in OPPOSITE directions: semantic identity makes *zero* locks — it fails closed — while coordinate identity keeps confidently locking wrong cells — it fails open. This asymmetry, not raw accuracy, is the deepest argument for meaning-based identity: when the world does not support identification, the semantic system knows it.

6.1 The full stack, and removing the last axis

Online alignment in the peer stack. With $N=4$ agents under scarcity, all in distinct secret frames, alignment develops DURING the episode: agents explore, co-visited landmarks accumulate, and per-sender frames lock in mid-run (onset in 100% of episodes, mean tick ≈ 15). Pooled success: shared-frame oracle 0.456, coordinate under misalignment 0.350, semantic 0.444 — **88% of the gap recovered**, with strict W^* functioning through recovered frames.

The law without the frame. Re-running the companion distance law with the trust $\times$ staleness gate ON TOP of semantic identity under a frame offset gives breakpoints 20/14/9 — *the same integers as the shared-frame closed form, exactly*. The gate and the frame recovery are orthogonal layers: the law does not notice that the shared grid is gone.

SE(2): no shared north. With secret rotations added, one rotation hypothesis per 90° reuses the translation aligner and the winner must *dominate* ($\geq 2\times$ the runner-up’s consensus). Results: exact (r^*, δ^*) recovery in **80/80** rotation $\times$ offset $\times$ seed combinations with 100% strict- W^* completion; in a 180° -symmetric world — where the frame is unrecoverable *in principle* — the aligner refuses (0/10 locks) while coordinates fail open (10/10 wrong locks); and the distance law again does not notice (bp = 20 under a 90° -rotated witness). The translation-only aligner, for contrast, never recovers a rotated frame (0/60) but does not always fail closed: in 2/60 cells a 180° -symmetric hazard substructure lets it fail OPEN — the empirical motivation for the dominance rule.

The corner trap. In the full stack, raw success is non-discriminating under rotations (rotated coordinate poison mostly maps off-grid and degrades into pseudo-exploration), so the registered discriminator is *warp-lock precision*: the share of strict W^* locks pointing at real water. Before the void filter, early border fingerprints (agents start in corners; rectangle borders are 180° -symmetric) fed wrong-frame hypotheses and semantic precision was 0.159; with borders excluded as non-landmarks it is 1.000 (8/8) **vs.** 0.000 (0/992 **phantom locks**) for coordinates, with alignment onset in 100% of episodes and clean regressions on all earlier experiments (Figure 3).

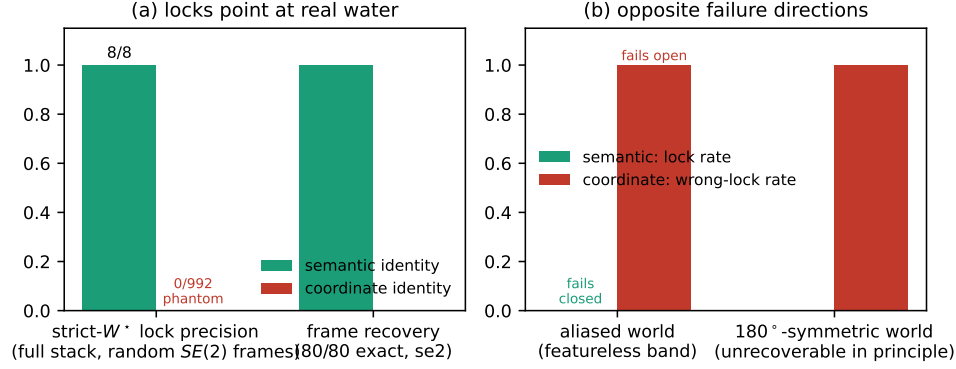


Figure 3: Meaning-based identity under $SE(2)$ frames. **(a)** Every strict- W^* lock the semantic stack makes points at real water (8/8); the coordinate stack makes 992 locks, all phantoms (0/992); frame recovery itself is exact in 80/80 witness-traveler combinations. **(b)** The failure asymmetry in worlds where identification is impossible (featureless band; a 180° -symmetric layout where the frame is unrecoverable in principle): semantic identity refuses to lock — fails closed — while coordinate identity keeps confidently locking wrong cells — fails open.

7 Related work

Map merging and place recognition. Multi-robot SLAM treats inter-map data association as an estimation problem over metric features Cadena et al. (2016); visual place recognition Lowry et al. (2016) matches appearance. Our contribution is not a competing matcher but the measurement layer on top: provenance-conditioned completion through recovered frames, the fail-closed/fail-open asymmetry as a registered prediction, and law invariance under frame removal.

Route sharing. Stigmergy transfers paths anonymously (pheromone gradients); learned communication (e.g. CommNet Sukhbaatar et al. (2016)) transfers policies opaquely. Neither exposes an edge-provenance decomposition, so neither can state — let alone predict — where a transferred route breaks. The rupture predictor is Time Warp Jefferson (1985) thinking applied to memory decay: deterministic replay of a gate over calibrated kinematics.

Songlines. The narrative frame is now load-bearing in both directions: the song is a sequence (route warp), its places are recognised by meaning (semantic identity), it fades verse by verse (rupture law), and it protects the walker while it lasts (hazard stratification).

8 Limitations and honest negatives

(1) Grid worlds; translation and 90° rotations only — continuous $SE(2)$ and appearance noise are future work (the fingerprint threshold is exact-equality because the substrate is noise-free; noisy substrates need the margin machinery of place recognition). (2) Four registered predictions failed and are reported with mechanisms: the $D=1$ negative control in ticks (turn economy; exact in path length), gain monotonicity in D (censoring; the truth is a cliff), the full-stack success comparison under rotations (non-discriminating; replaced by lock precision), and “translation-fails-closed” (2/60 fail open via symmetric substructures). (3) The route follower is a new planner tier; all three arms share it, but route-warp value is measured relative to this consumer. (4) Hazard safety transfers by construction when the witness’s route is safe; the measured content is that place transfer CANNOT carry it and that the protection ends at the predicted rupture. (5) Scarcity-level completion of warp locks remains governed by the collision phenomenology of the companion paper.

9 Conclusion

Strip the song down to a pin on a shared map and two things break invisibly: the HOW (fatal at the first wall, expensive in every hazard field) and the WHO-MEANS-WHAT (fatal under any

197 frame disagreement, and dangerous precisely because it fails open). Restore the song — edges
 198 with provenance, places with meaning — and both failures become measurable, lawful, and largely
 199 repaired: routes transfer completion and safety and break only where the aging law says; meaning
 200 recovers frames exactly, refuses honestly when the world is ambiguous, and carries the entire
 201 measurement stack — φ , gates, distance law — unchanged into a world where agents share no
 202 coordinates at all. The pin was never the memory. The song is.

203 **A Registered predictions and revisions**

204 All predictions were written to disk before execution; the artifact retains both versions wherever an
 205 operationalisation was revised. Revisions: R1 negative control re-measured in path length; R1 gain
 206 monotonicity re-formulated as cliff detection; W9 full-stack discriminator changed from success to
 207 warp-lock precision; W8 strict- W^* functioning extended to include warp-assisted own completions
 208 (taxi chains).

209 **B Reproducibility**

```
210 PYTHONPATH=. python experiments/warp/exp_route_warp_r0.py
211 PYTHONPATH=. python experiments/warp/exp_route_warp_r1.py --seeds 20
212 PYTHONPATH=. python experiments/warp/exp_route_warp_r2.py --seeds 5
213 PYTHONPATH=. python experiments/warp/exp_route_warp_r3.py --seeds 20
214 PYTHONPATH=. python experiments/warp/exp_warp_semantic_identity.py
215 PYTHONPATH=. python experiments/warp/exp_warp_semantic_stack.py --seeds 20
216 PYTHONPATH=. python experiments/warp/exp_warp_rotation.py --seeds 10
```

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